A (More) Objective View of AI

Software Development in the Age of AI

Federico Busato

2026-04-13

1. **Software Employment**

2. **Limitations of AI-Generated Code**
   - 2.1 Slot Machine and the Illusion of Control
   - 2.2 Generalization
   - 2.3 Creativity
   - 2.4 The Illusion of Competence
   - 2.5 Technical Debt
   - 2.6 Security Vulnerabilities
   - 2.7 The “Last Mile” Problem

3. **AI as a Software Engineering Tool**
   3.1 Code Generation is NOT Software Engineering
   3.2 The Role of Human Expertise

4. **Engineering Practices in the Age of AI**

1. Software Employment

AI Replacing Developers Misconception

Despite the daily dystopian and gloomy predictions on social media and news sites, AI will not replace software developers. In fact, demand for software developers is and will remain strong.

Layoffs and the decline in entry-level job postings have been driven by **massive capital spending on infrastructure** and economic uncertainty, disproportionately affecting early-career roles. *The industry is effectively trading junior headcount for compute capacity.* [1, 2, 3]

[1] Why AI Companies May Invest More than $500 Billion in 2026 - Goldman Sachs

[2] AI isn't replacing jobs. AI spending is - Fast Company, 2025

[3] The cost of compute: A $7 trillion race to scale data centers - McKinsey & Company, 2025

AI Replacing Developers Misconception 2/2

A Harvard Business School study [4] found that at firms adopting AI, junior employment drops significantly while senior employment remains flat. The researchers warn that this is a *"defensive move"* that boosts short-term efficiency but **risks a long-term talent crisis**, as fewer human experts will be trained to supervise the AI systems of the future.

[4] Generative AI as Seniority-Biased Technological Change - Social Science Research Network, 2025

*"The impact of productivity gains on jobs depends on whether the demand for goods produced by that work is **elastic** or **inelastic**. If demand is inelastic, productivity gains mean jobs will be lost...*

... If product demand is sufficiently elastic, productivity-enhancing technology will increase industry employment" [5]

— **Shaping AI's Impact on Billions of Lives**

[5] Shaping AI's Impact on Billions of Lives - arXiv, 2025 / Communications of the ACM, 2026

AI and Software Employment

Software development has elastic demand. It offers great potential and opportunities, driven by its integration into every aspect of life and business and by rapid technological evolution.

- The US Bureau of Labor Statistics predicts a **25% increase** in software developers over the next decade [6].
- The World Economic Forum lists software and application developers among the **fastest-growing jobs** in the 2025-2030 timeframe [7].
- The global software market size is projected to reach $1.4 trillion by 2030, growing at a compound annual **growth rate** (CAGR) of **+11.3%** [8].

---

[6] Employment Projections, 2025-2035 - US Bureau of Labor Statistics, 2025

[7] Future of Jobs Report, 2025-2030 - World Economic Forum, 2025

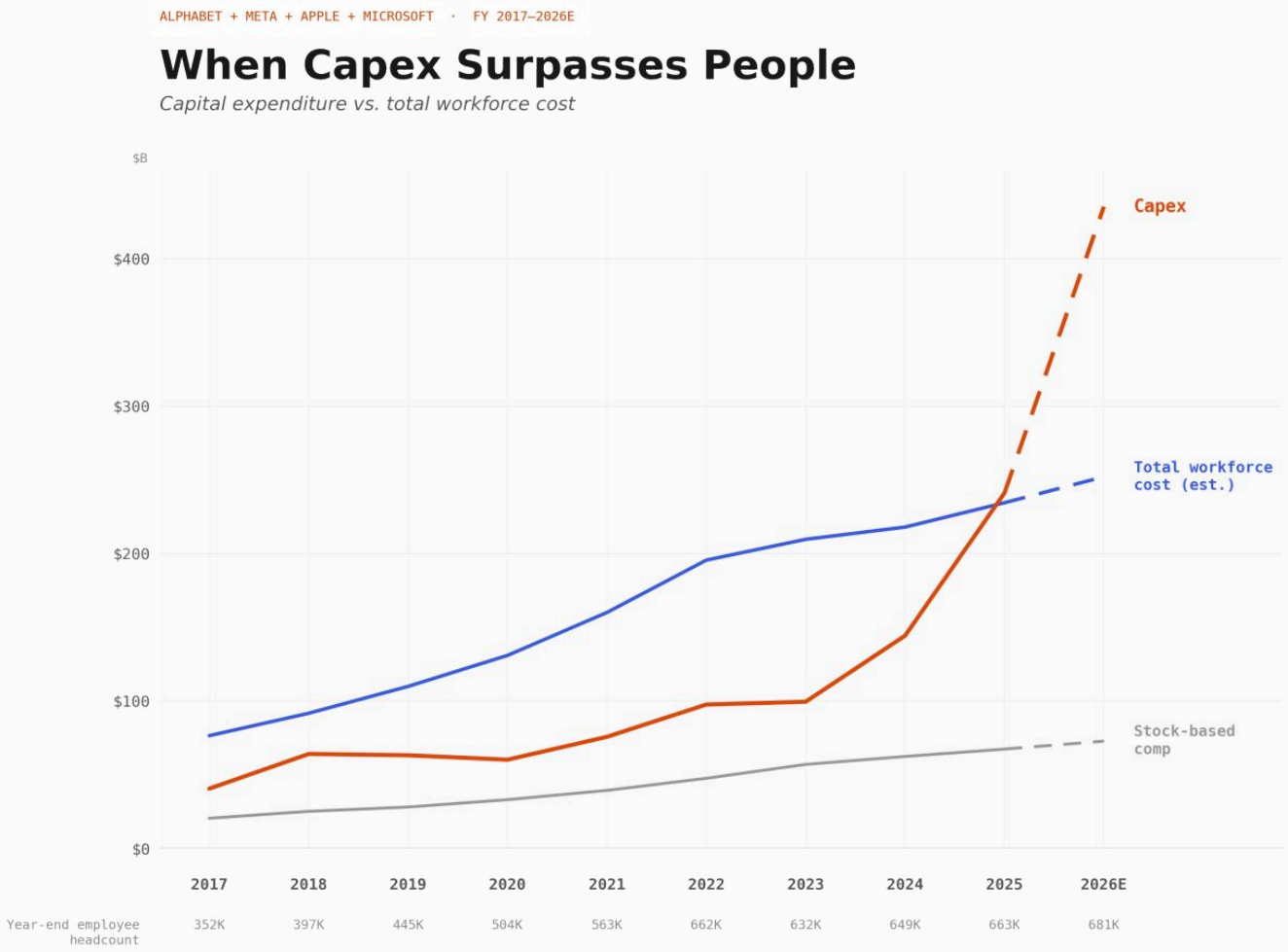
[8] Software Market (2025 - 2030) - Grand View Research, 2025

- Engineering headcount changes since 2022: Meta **+19%**, Google **+16%**, Apple **+13%** [9].

> ~ *Jobs with high exposure to AI have seen greater wage and stronger job growth than jobs minimally exposed to AI* [10].
>
> — **AI exuberance: Economic upside, stock market downside**

[9] Software Engineer Job Market 2025 - Underdog.io, 2025

[10] AI exuberance: Economic upside, stock market downside - Vanguard economic and market outlook for 2026

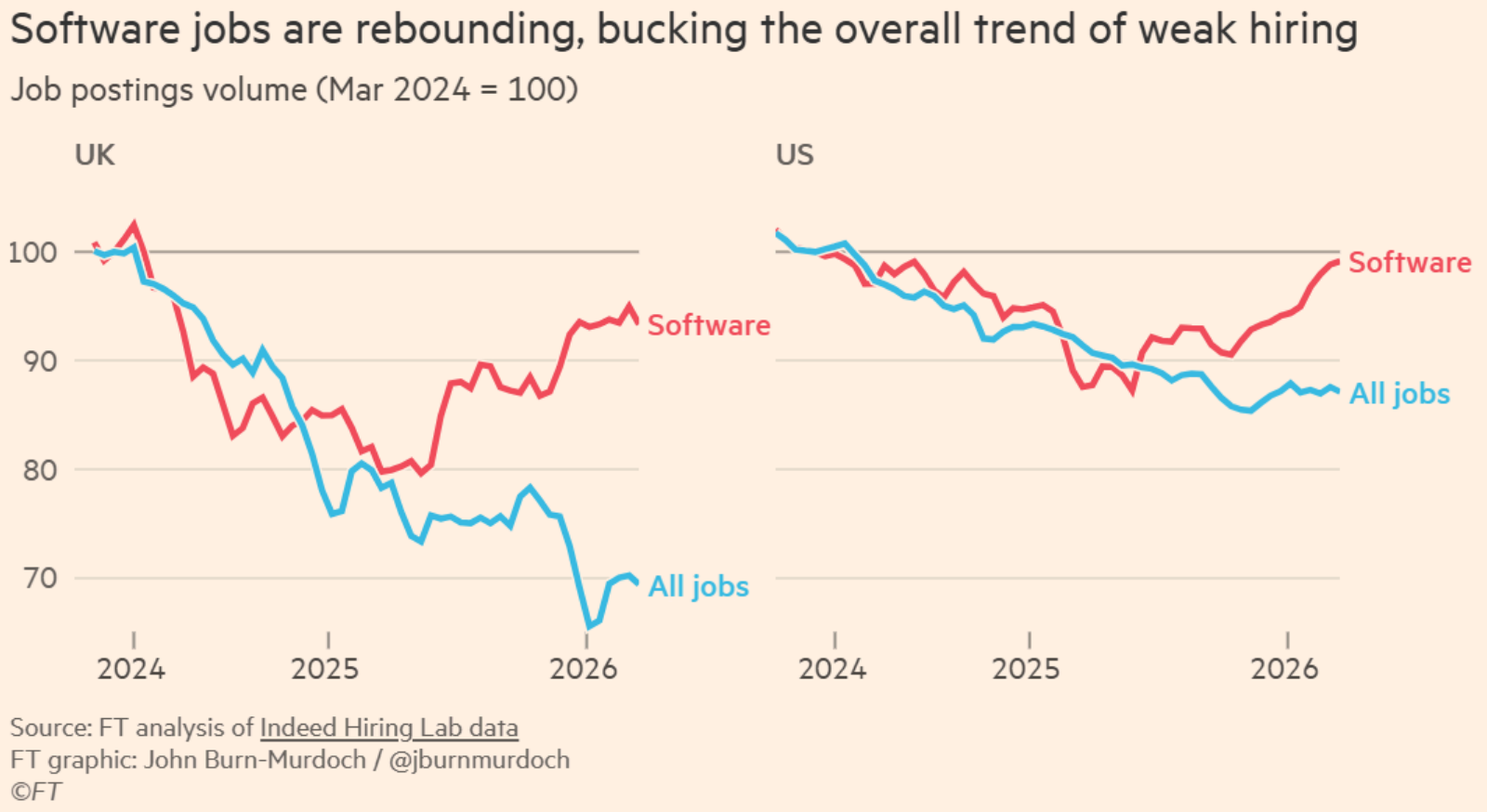


[11] When Capex Surpasses People - Prasad Setty, 2026

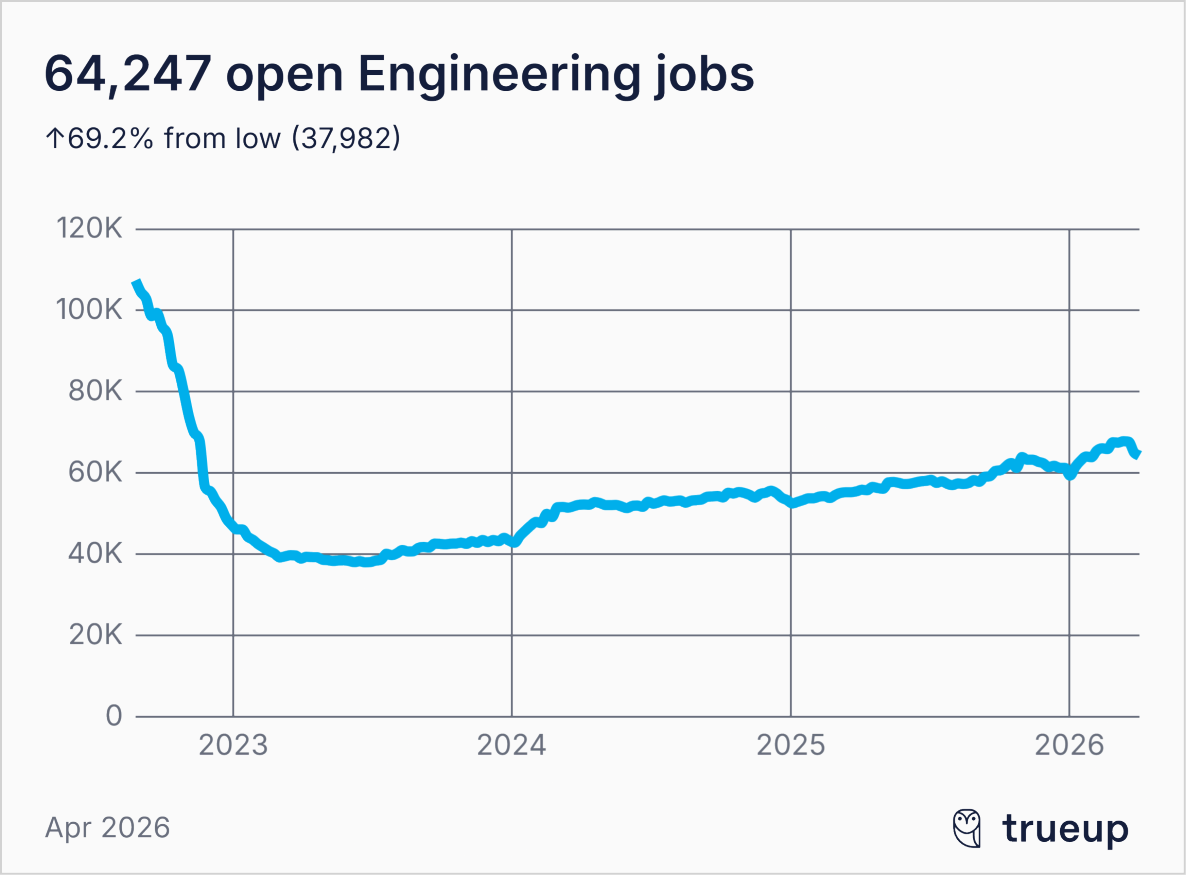
Occupations with the highest projected total change in employment, 2024-2034 [12]

Occupation	Projected new jobs	Median annual pay, 2024
Home health and personal care aides	739.8K	$34.9K
Software developers	267.7K	$133.1K
Stockers and order fillers	235K	$37.1K
Fast food and counter workers	233.2K	$30.5K
Cooks, restaurant	217K	$36.8K
Registered nurses	166.1K	$93.6K
General and operations managers	164K	$103K
Medical and health services managers	142.9K	$118K
Financial managers	128.8K	$161.7K
Nurse practitioners	128.4K	$129.2K
Construction laborers	106.5K	$46.7K
Computer and information systems managers	101.6K	$171.2K

[12] What are the fastest-growing professions in America? - USAFacts, 2025


Software jobs are rebounding, bucking the overall trend of weak hiring
Job postings volume (Mar 2024 = 100)
UK
US
100
90
80
70
Software
All jobs
Software
All jobs
2024
2025
2026
2024
2025
2026
Source: FT analysis of Indeed Hiring Lab data
FT graphic: John Burn-Murdoch / @jburnmurdoch
©FT

[13] The AI Shift: Will software engineers survive agentic AI? - Sarah & Murdoch, Financial Times, 2026



[14] Engineering Job Report, April 2026 - TrueUp, 2026

"Without the hiring of early-in-career developers, the profession's talent pipeline will collapse, and organizations will face a future without the next generation of experienced engineers." [15]

— **Redefining the Software Engineering Profession for AI**,
Communications of the ACM, 2026

[15] Redefining the Software Engineering Profession for AI - Russinovich & Hanselman, Communications of the ACM, 2026

"Now that we have software that can write working code, ***what is there left for us humans to do?***

The answer is so much stuff.

Writing code has never been the sole activity of a software engineer. The craft has always been figuring out what code to write. Any given software problem has dozens of potential solutions, each with their own tradeoffs. Our job is to navigate those options and find the ones that are the best fit for our unique set of circumstances and requirements." [16]

— **What is agentic engineering?**,
S. Willison, 2026

[16] What is agentic engineering? - S. Willison, 2026

2. Limitations of AI-Generated Code

Slot Machine and the Illusion of Control

*"The thing about AI based coding is that it's like a **slot machine**, and that you you have an **illusion of control**, you know, you can get to craft your prompt, and your list of MCPs, and your skills, and whatever, and then in the end, you pull the lever. Right?*

*... **It's the stochastic thing. You get the occasional win. It's like, oh, I won. I got a feature.**"* [17]

— **The Dangerous Illusion of AI Coding?**,
Jeremy Howard, 2026

[17] The Dangerous Illusion of AI Coding? - Jeremy Howard, 2026

Generalization

- **Lack of generalization beyond the training data.**

> *"This brittleness means that* ***coding agents are almost frighteningly good at what they've been trained and fine-tuned on—modern programming languages****, JavaScript, HTML, and similar well-represented technologies—and generally* ***terrible at tasks on which they have not been deeply trained****.*
>
> *... It took me five minutes to make a nice HTML5 demo with Claude but a week of torturous trial and error, plus actual systematic design on my part, to make a similar demo of an Atari 800 game."* [32]

— **Benj Edwards**,
Ars Technica

Creativity 💬

"Due to what might poetically be called "preconceived notions" baked into a coding model's neural network (more technically, statistical semantic associations), ***it can be difficult to get AI agents to create truly novel things****, even if you carefully spell out what you want."* [32]

— **Benj Edwards**,
Ars Technica

- **Bug proliferation**

> *"**Fixing bugs can also create bugs elsewhere. Coding agents supercharge this phenomenon** because they can barrel through your code and make sweeping changes in pursuit of narrow-minded goals that affect lots of working systems."* [32]
>
> — **Benj Edwards**, Ars Technica

> *"A comprehensive report analyzed 470 real-world open-source pull requests and found that **AI-generated code introduces 1.7x more defects** across every major category of software quality—including logic, maintainability, security, and performance."* [18]
>
> — **CodeRabbit**

[18] State of the AI vs. Human Code Generation Report - CodeRabbit, 2025

*"**LLMs cosplay understanding things.***

*... **The difference between pretending to be intelligent and actually being intelligent is entirely unimportant, as long as you're in the region in which the pretense is actually effective**, you know. So it's actually fine for a great many tasks that LLMs only pretend to be intelligent, because for all intents and purposes, it just doesn't matter **until you get to the point where it can't pretend anymore**. And then you realize, like, oh my god. This thing is so stupid."* [17]

— **Jeremy Howard**,
The Dangerous Illusion of AI Coding?, 2026

The Illusion of Competence

"61% agree that "AI often produces code that looks correct but isn't reliable." That's a critical finding—it means ***AI code can introduce subtle bugs that are harder to spot than typical human errors.***

*The same percentage (61%) agree that it "****requires a lot of effort to get good code from AI****" through prompting and fixing."* [19]

— **Sonar**,
State of Code Developer Survey report, 2026

[19] State of Code Developer Survey report - Sonar, 2026

- **AI-assisted code has contributed to a measurable decline in code quality.** Code generated by LLMs without engineering supervision tends to deteriorate quickly over time due to duplicated logic, inconsistent naming, and poor architectural organization. This translates into rising maintenance costs, lower developer satisfaction, and fragile systems.

My project became so big that claude can't properly understand it

Question

So, I made a project in python entirely using Cursor (composer) and Claude, but it has gotten to a point that the whole codebase is over 30 Python files, code is super disorganized, might even have duplicate loops, and Claude keeps forgetting basic stuff like imports at this point. When I ask it to optimize the code or to fix a bug, it doesn't even recognize the main issue and just ends up deleting random lines or breaking everything completely.

I have 0 knowledge about python, it's actually a

*"**The share of copy/pasted lines surged** from 8.3% in 2020 to 12.3% in 2024, a 48% relative increase.*

*The proportion of new code that was revised within two weeks of its initial commit grew from 3.1% in 2020 to 5.7% in 2024, indicating a **rise in premature or low-quality commits**.*

*The percentage of revised code that was originally written more than a month prior dropped from 30% in 2020 to just 20% in 2024. This suggests that **developers are spending more time modifying recently written AI-generated code rather than improving or refactoring** legacy systems."* [20]

— **GitClear**,
AI Copilot Code Quality, 2025

[20] AI Copilot Code Quality: 2025 Look Back at 12 Months of Data - GitClear, 2025

*"**AI coding tools have caused as many problems as they have solved**, according to industry experts. The easy-to-use and accessible nature of AI coding tools has enabled a flood of bad code that threatens to overwhelm projects. **Building new features is easier than ever, but maintaining them is just as hard and threatens to further fragment software ecosystems.**"*

... Blender Foundation CEO Francesco Siddi said LLM-assisted contributions typically 'wasted reviewers' time and affected their motivation.

... The open source data transfer program cURL recently halted its bug bounty program after being overwhelmed by what creator Daniel Stenberg described as 'AI slop.'" [21]

— **For open source programs, AI coding tools are a mixed blessing**, TechCrunch, 2026

[21] For open source programs, AI coding tools are a mixed blessing - R. Brandom, TechCrunch, 2026

"In the 1980 Turing Award lecture Tony Hoare said: 'There are two ways of constructing a software design: one way is to make it so simple that there are obviously no deficiencies, and the other is to make it so complicated that there are no obvious deficiencies.' This LLM-generated code falls into the second category. The reimplementation is 576,000 lines of Rust. ***That is 3.7x more code*** *than SQLite."* [35]

— **Your LLM Doesn't Write Correct Code. It Writes Plausible Code.**, Hōrōshi, 2026

Security Vulnerabilities

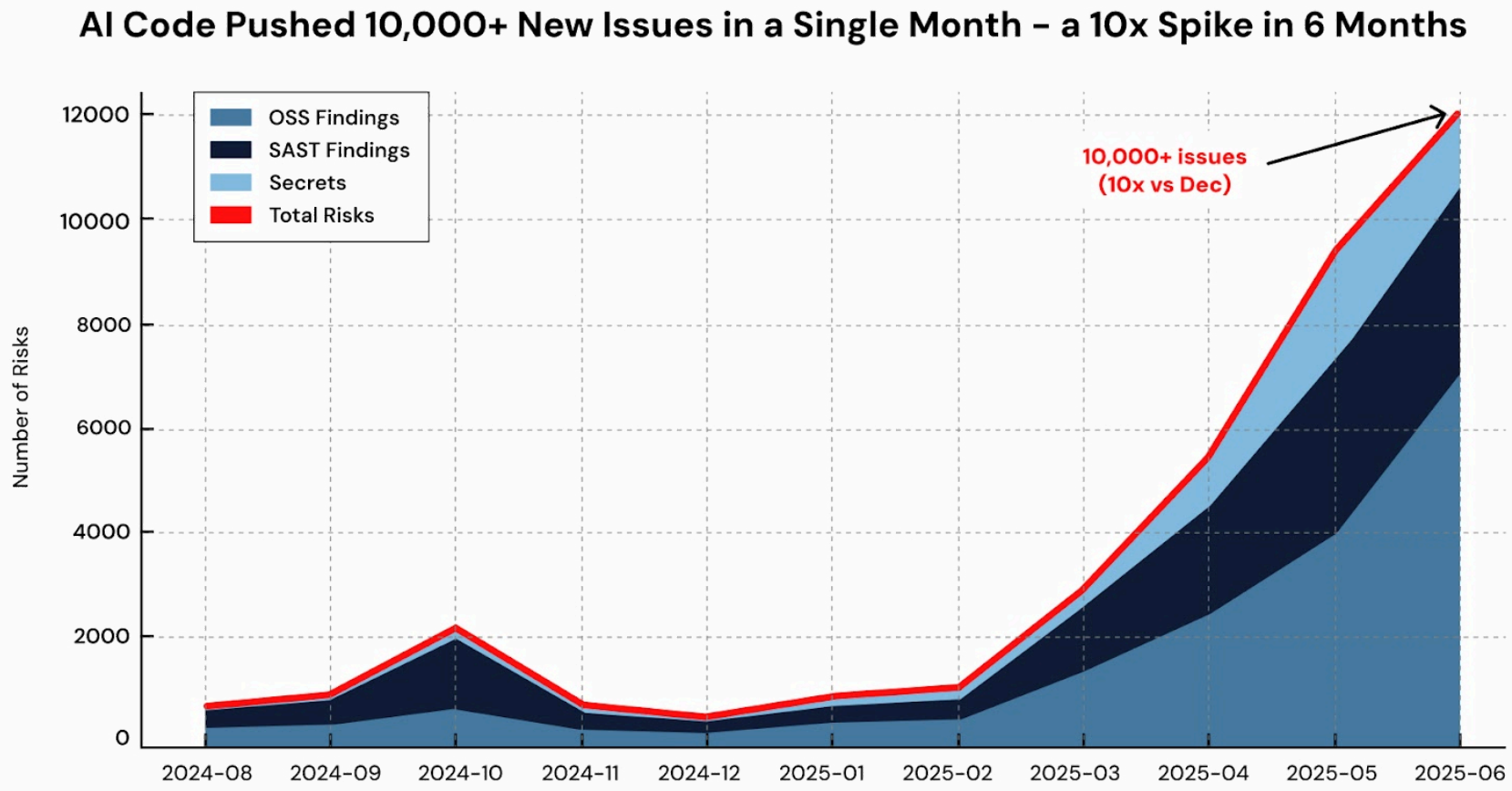
- **AI-generated code can introduce real risks for security and reliability.**

> *"We produce 89 different scenarios for Copilot to complete, producing 1,689 programs. Of these, **we found approximately 40% to be vulnerable**."* [22]
>
> — **Asleep at the Keyboard?**, Pearce et al.

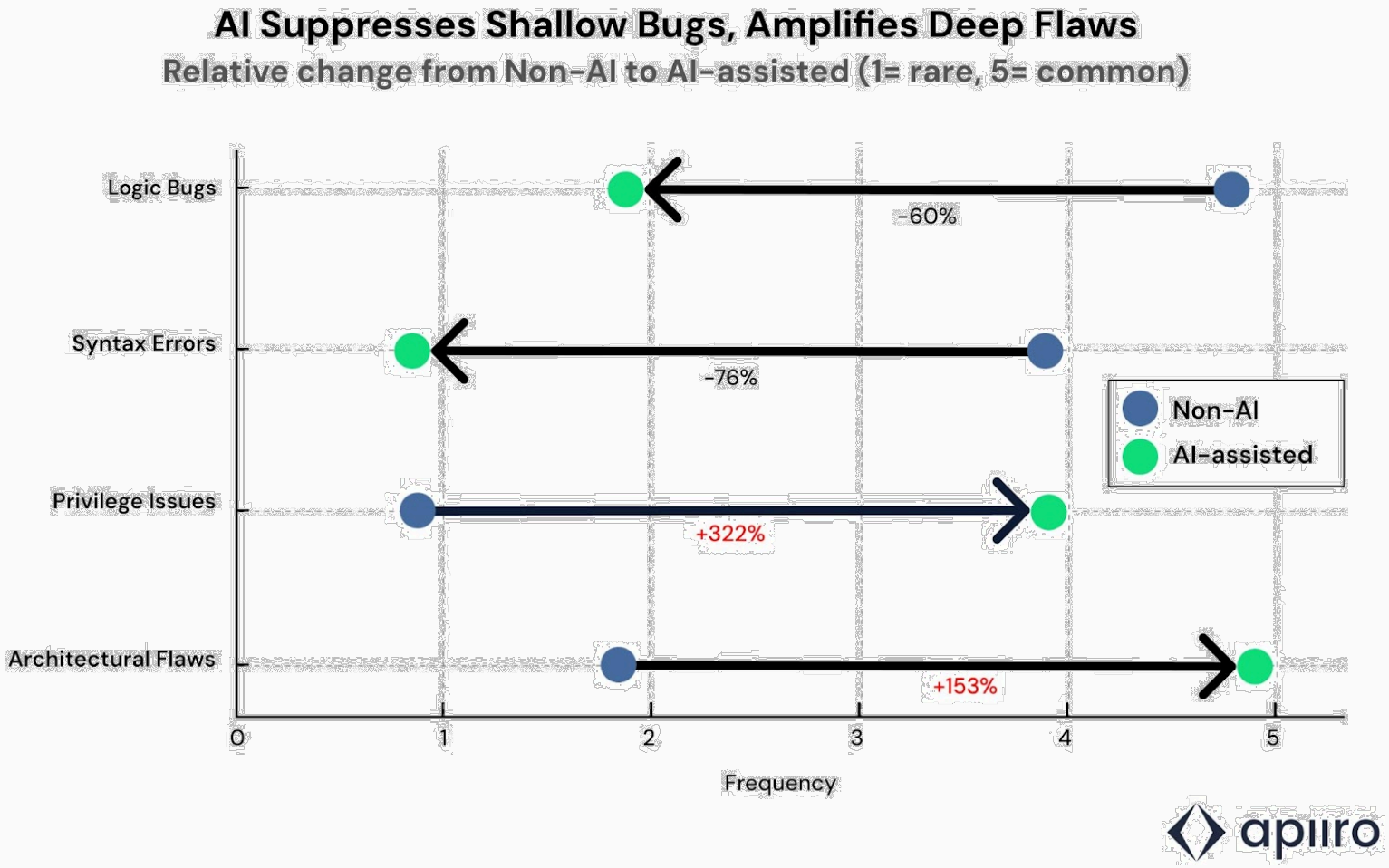
> *By June 2025, AI-generated code was introducing over 10,000 new security findings per month across the repositories in our study—a 10x spike in just six months compared to December 2024. And **the curve isn't flattening; it's accelerating**.* [23]
>
> — **4x Velocity, 10x Vulnerabilities**, Apiiro

[22] Asleep at the Keyboard? Assessing the Security of GitHub Copilot's Code Contributions - Pearce et al., Communications of the ACM, 2025

[23] 4x Velocity, 10x Vulnerabilities: AI Coding Assistants Are Shipping More Risks - Apiiro, 2025


AI Code Pushed 10,000+ New Issues in a Single Month – a 10x Spike in 6 Months
OSS Findings
SAST Findings
Secrets
Total Risks
10,000+ issues
(10x vs Dec)
Number of Risks
12000
10000
8000
6000
4000
2000
0
2024-08
2024-09
2024-10
2024-11
2024-12
2025-01
2025-02
2025-03
2025-04
2025-05
2025-06

apiiro


AI Suppresses Shallow Bugs, Amplifies Deep Flaws
Relative change from Non-AI to AI-assisted (1= rare, 5= common)
Logic Bugs
Syntax Errors
Privilege Issues
Architectural Flaws
-60%
-76%
+322%
+153%
Non-AI
AI-assisted
0
1
2
3
4
5
Frequency
apiiro

> *"We are still early, and **fully autonomous development comes with real risks**. When a human sits with Claude during development, they can ensure consistent quality and catch errors in real time. For autonomous systems, **it is easy to see tests pass and assume the job is done, when this is rarely the case**."* [24]
>
> — **Building a C compiler with a team of parallel Claudes**, Anthropic, 2026

[24] Building a C compiler with a team of parallel Claudes - Anthropic, 2026

The “Last Mile” Problem 1/3

- **The “Last Mile” Problem**. Prototyping with AI is not the same as building a production-quality product.

> *“The first 90 percent of an AI coding project comes in fast and amazes you.* ***The last 10 percent involves tediously filling in the details*** *through back-and-forth trial-and-error conversation with the agent.”* [32]
>
> — **Benj Edwards**,
> Ars Technica, 2026

The "Last Mile" Problem 💬

"Honest reflections from coding with AI so far as a non-engineer:

It can get you 70% of the way there, but that ***last 30% is frustrating****. It keeps taking one step forward and two steps backward with new bugs, issues, etc.*

If I knew how the code worked I could probably fix it myself. But since I don't, ***I question if I'm actually learning that much.****"* [25]

— **Peter Yang**

[25] Peter Yang

The “Last Mile” Problem 💬

With AI, it’s easier to get the first 90 percent out there. This means we can spend more time on the remaining 10 percent, which means more time for craftsmanship and figuring out how to make your users happy. [26]

— **The 100 hour gap between a vibecoded prototype and a working product**, M. Budkowski, 2026

[26] The 100 hour gap between a vibecoded prototype and a working product - M. Budkowski, 2026

3. AI as a Software Engineering Tool

Code Generation is NOT Software Engineering 1/6

- **LLMs should not be seen as a substitute for software engineering.**

> ***Software engineering is a unusual discipline, and a lot of people mistake it for being the same as typing code into an IDE.***
>
> *... Because software engineering is all about finding what those pieces are, and how they should behave, and then how you can put them together to create a bigger piece, and then how you can put them together to create a bigger piece.* ***And if we do that well, then in 10 years time, we could have software that is far more capable than anything we could even imagine today.***" [27]
>
> — **Jeremy Howard**,
> The Dangerous Illusion of AI Coding?, 2026

[27] The Dangerous Illusion of AI Coding? - Jeremy Howard, 2026

*"**One of the lesser spoken truths of working productively with LLMs as a software engineer on non-toy-projects is that it's difficult**. There's a lot of depth to understanding how to use the tools, there are plenty of traps to avoid, and the pace at which they can churn out working code raises the bar for what the human participant can and should be contributing.*

*... Iterating with coding agents to **produce production-quality** code that I'm confident **I can maintain in the future** feels like a different process entirely."* [28]

— **Simon Willison**,
Vibe engineering

[28] Vibe engineering - Simon Willison, 2025

> *I'm not very happy with the code quality and I think* ***agents bloat abstractions****, have poor code aesthetics, are very prone to copy pasting code blocks and* ***it's a mess****, but at this point I stopped fighting it too hard and just moved on.* [29]
>
> — **Andrej Karpathy**

[29] Andrej Karpathy, 2026

*"**Programming, like writing, is an activity, where one iteratively sharpens what they're doing as they do it.***

*... But, **vibe coding gives the illusion that your vibes are precise abstractions**. They will feel this way right up until they leak, which will happen when you add enough features or get enough scale. **Unexpected behaviors (bugs) that emerge from lower levels of abstraction that you don't understand** will sneak up on you and wreck your whole day."* [30]

— **Reports of code's death are greatly exaggerated**,
Steve Krouse

[30] Reports of code's death are greatly exaggerated - Steve Krouse, 2026

*"**LLMs** have effectively killed the cost of generating lines of code, but they **haven't touched the cost of truly understanding a problem**. We're seeing a flood of "apps built in a weekend," but most of these are just thin wrappers around third-party APIs. They look impressive in a Twitter demo, but they often crumble the moment they hit the friction of the real world...*

*... In this new reality, **engineering expertise remains incredibly valuable**, but the nature of the role is shifting. Relevance is not fading. Instead, it is about leveraging these tools to build at a higher level than was previously possible. True expertise is now required to steer these systems and provide the technical oversight that LLMs currently lack."* [31]

— **Code Is Cheap Now. Software Isn't**,
Chris Gregori

[31] Code Is Cheap Now. Software Isn't - Chris Gregori, 2026

"Creating durable production code, managing a complex project, or crafting something truly novel still requires experience, patience, and skill beyond what today's AI agents can provide on their own.

... veteran software developers probably shouldn't fear losing their jobs to these tools any time soon. ***In fact, they may become busier than ever****.*

... I don't think AI tools will make human software designers obsolete. Instead, they may well help those designers become more capable." [32]

— **Benj Edwards**,
Ars Technica

[32] 10 things I learned from burning myself out with AI coding agents - Benj Edwards, Ars Technica, 2026

- **AI systems complement human expertise. They do not substitute for it.**

> *"What the models do, the capabilities they have, and the quality of the content they produce is, to a large extent,* ***a reflection of the user****."* [33]
>
> — **The skeptic's guide to generative AI assisted coding**, Rob Patro

[33] The skeptic's guide to generative AI assisted coding - Rob Patro, combine-lab, 2026

*"**Experienced human software developers bring judgment, creativity, and domain knowledge that AI models lack**. They know how to architect systems for long-term maintainability, how to balance technical debt against feature velocity, and when to push back when requirements don't make sense.*

*... They can automate many tasks, but **managing the overarching project scope still falls to the person telling the tool what to do**."* [32]

— **Benj Edwards**,
Ars Technica, 2026

"Knowing something about how good software development works helps a lot when guiding an ***AI coding agent—the tool amplifies your existing knowledge rather than replacing it****."* [32]

— **Benj Edwards**, Ars Technica

*"****AI tools amplify existing expertise****. The more skills and experience you have as a software engineer the faster and better the results you can get from working with LLMs and coding agents."* [28]

— **Simon Willison**, Vibe engineering

"Relying too much on AI risks missing subtle bugs, architectural flaws, and vulnerabilities that only skilled engineers can catch. ***Human oversight, critical thinking, and domain knowledge are indispensable*** *for both correcting errors and driving innovation as technology progresses.*

Generative AI currently acts as seniority-biased technological change: ***It disproportionately amplifies engineers who already possess systems judgment****, like taste for architecture, debugging under uncertainty, and operational intuition."* [15]

— **Redefining the Software Engineering Profession for AI**,
Communications of the ACM, 2026

The Role of Human Expertise 💬

"When I was working on something I already understood deeply, ***AI was excellent****. I could review its output instantly, catch mistakes before they landed and move at a pace I'd never have managed alone.*

... When I was working on something I could describe but didn't yet know, ***AI was good but required more care****.*

... When I was working on something where I didn't even know what I wanted, ***AI was somewhere between unhelpful and harmful****.*

... But expertise alone isn't enough. Even when I understood a problem deeply, ***AI still struggled if the task had no objectively checkable answer*** [34]

— **Eight years of wanting, three months of building with AI**,
L. Maganti, 2026

[34] Eight years of wanting, three months of building with AI - L. Maganti, 2026

Plausibility over Correctness

*"**LLMs optimize for plausibility over correctness**. In this case, plausible is about 20,000 times slower than correct.*

*... THIS is the failure mode. Not broken syntax or missing semicolons. The code is syntactically and semantically correct. **It does what was asked for. It just does not do what the situation requires.**"* [35]

— **Your LLM Doesn't Write Correct Code. It Writes Plausible Code.**, Hōrōshi, 2026

[35] Your LLM Doesn't Write Correct Code. It Writes Plausible Code. - Hōrōshi, 2026

4. Engineering Practices in the Age of AI

Engineering Practices in the Age of AI

LLMs reward existing software engineering practices. The following list presents a concise set of practices through which AI systems can enhance the software development process. This list has been compiled from engineers' public notes, discussions with colleagues, and personal experience.

- **Code Review**. LLMs rely on pattern matching and memorization. They can be very effective at finding common issues, faulty logic, or missing assumptions in code. *This is not a substitute for human code review*.

- **Documentation**. Writing documentation is essential for understanding and maintaining software, but it is often a time-consuming and boring task. LLMs can help write documentation, or a draft, quickly, *to review and refine later*.

- **Code Explanation**. LLMs can parse code much faster than humans. As a result, they can be very useful for explaining code in context or for understanding the organization and workflow of large codebases.

- **Prototyping**. Productization is the most demanding phase of the development process, and it requires a clear idea of what to achieve, which is often not the case. LLMs can be very effective at quickly implementing new features or ideas, *even if the quality is far from production-ready*.

- **Discussing Ideas and Intuitions**. It is common to reflect on new ideas or revisit earlier ones when thinking through specific aspects of a problem. LLMs can discuss these topics much as a coworker might, even before prototyping.

- **Evaluating Alternatives**. There are dozens of ways to solve a given problem. LLMs can suggest alternatives that users can evaluate and then select the most suitable option *depending on the context*.

Engineering Practices in the Age of AI 3/5

- **Enforcing Coding Style**. Large codebases involving several engineers tend to develop high-level practices that are often difficult to enforce with common tools. Coding style can fall into this category. Requirements such as "the code shall not use lambda expressions" are difficult to enforce with traditional tools but trivial for LLMs.

- **Testing**. Robust and comprehensive test suites prevent users *and LLMs* from unintentionally introducing bugs. Additionally, LLMs can quickly draft tests that can be adjusted later.

- **Refactoring**. Technology evolves faster than engineers can keep up with, especially in projects with limited engineering resources. LLMs are an excellent tool for modernizing codebases, but it is important to pay attention to *maintaining the underlying logic*.

- **Debugging**. LLMs can automate the debugging process, especially for simple bugs. They can compile, execute, analyze the output, or even navigate the `git` history and find connections in the code to identify problems or formulate hypotheses.

- **Vibe Coding Tools**. It is common to face limitations with open-source tools because each project has its own specific context. Engineering work often does not leave time to contribute to external projects to overcome these limitations. LLMs can implement small features while engineers stay focused on the actual code. One example could be adding a new check to `clang-tidy`.

- **Syntax Helper**. It is common to work on projects written in programming languages outside our area of expertise. LLMs can assist by explaining new syntax and translating between languages.

- **Git Interaction**. LLMs show excellent capabilities when working with `git`. They can navigate history, reverse changes, and identify the root causes of bugs. They can also handle complex rebases without human intervention.